#3 If the Bloom filter returned `true`, we need to check if the server actually stores the contact, because it could be a false positive.
#4 Otherwise, since Bloom filters don’t have false negatives (but only false positives), we can return `false`.

For adding new contacts, instead, we always have to sync to our permanent storage[11], as shown in listing 4.3: since this likely imply a remote connection through a network, there is a non-negligible probability that the call to the server fails; therefore, we need to handle possible failures and make sure the remote call succeeds before also updating the Bloom filter.

To be thorough, in real implementations we should also synchronize the access to the server and Bloom filter, using a locking mechanism (see chapter 6), and use a try-catch around the whole operation, rolling back (or retrying) if the call on the Bloom filter fails.

Listing 4.3 Adding a new contact

```
function addContact(bloomFilter, server,
contact)                                #1
  if server.storeContact(contact)
then                                          #2

bloomFilter.insert(contact)
            #3
    return
true
    #4
  else
```